

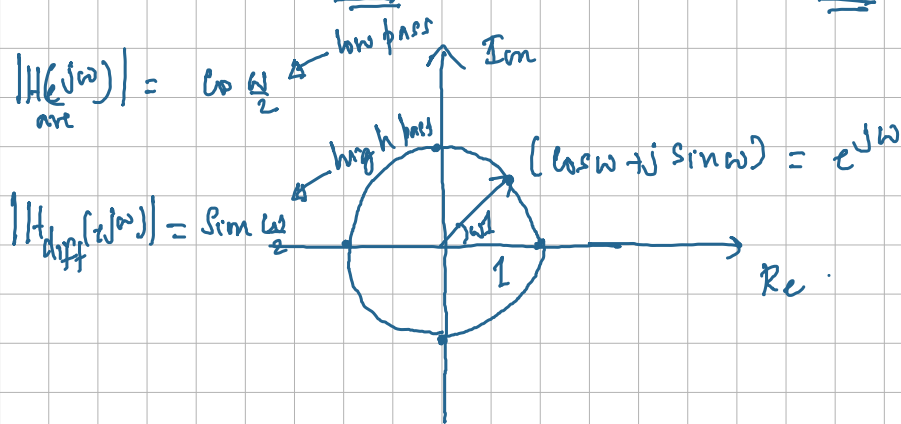
10 | 08 | 26

Signal Processing

Announcements

- Orig answers ✓
- Recap: DTFT and its utility
- Z-Transform
- Pause and reflect.

• DTFT: $H_{\text{ave}}(e^{j\omega}) = \frac{1+e^{-j\omega}}{2}$; $H_{\text{cliff}}(e^{j\omega}) = \frac{1-e^{-j\omega}}{2}$; $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$
 DTFT Analysis equation

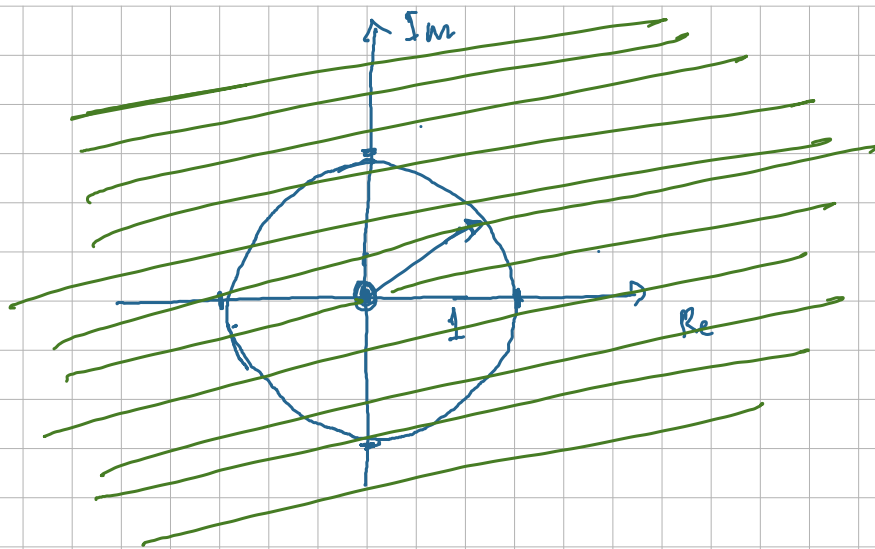


- Note that the DTFT is defined as an infinite sum. This naturally leads to questions related to convergence. E.g., if $x[n] = 1; -\infty < n < \infty$, what happens to its DTFT?
- To handle such cases and equip ourselves with more flexibility, we define the z-transform:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} ; \quad z = re^{j\omega} \rightarrow \begin{matrix} \text{radius} \\ \text{angle} \end{matrix}$$
- $H_{\text{ave}}(z) = \frac{1}{2}(1+z^{-1})$; $H_{\text{diff}}(z) = \frac{1-z^{-1}}{2}$. Both exist for all $z \neq 0$.

The region of convergence is the set of z for which the z -transform exists

- Let's put this all together.



$\parallel$: Z transform

②: DTFT

①: DFT